

Review

Using machine learning to predict the long-term performance of fibre-reinforced polymer structures: A state-of-the-art review

Chiara Machello^a, Milad Bazli^{a,b,*}, Ali Rajabipour^a, Hooman Mahdizadeh Rad^a, Mehrdad Arashpour^c, Ali Hadigheh^d^a Faculty of Science and Technology, Charles Darwin University, Darwin, Australia^b School of Mechanical and Mining Engineering, The University of Queensland, Queensland, Australia^c Department of Civil Engineering, Monash University, Australia^d School of Civil Engineering, The University of Sydney, Sydney, NSW 2006, Australia

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ABSTRACT

When exposed to environmental conditions, fibre-reinforced polymer (FRP) composites are prone to material degradation. The environmental reduction factor in different structural codes reflects the significant effect of the long-term durability of FRPs in aggressive environments. Traditional prediction methods rely on oversimplified premises, which may result in erroneous errors. Due to its proficiency in dealing with complex non-linear structural problems, machine learning (ML) offers a unique potential to increase the predictability of structural engineering factors. This can be attributed to the recent advancements in ML techniques, which leverage their robustness when handling large datasets, as well as the increased processing power that facilitates more efficient data analysis. This article reviews the current implementation cases and capabilities of ML algorithms in overcoming the shortcomings of conventional models for predicting the durability performance of FRP systems. According to the literature, it was found that the efficiency of ML approach varies significantly depending on the quality and comprehensiveness of the database. While various researcher-employed algorithms generally yield accurate predictions for retaining mechanical properties in FRP composites with minor errors, sensitivity analysis highlights varied impacts of variables when using different datasets or machine learning algorithms. This variance may arise from factors like inadequate or low-quality datasets, insufficient training, overfitting, and other influences. More experimental data are needed to enhance the current database to effectively apply ML in more applications for FRPs under different loading and environmental conditions. The paper ends by suggesting future research directions in this field.

1. Introduction

In engineering, using fibre-reinforced polymer (FRP) composite materials is becoming more common than ever [1–3]. FRPs are a competitive alternative to traditional materials like concrete and steel since they are known for having exceptional strength- and stiffness-to-weight ratios [4–6]. Additionally, because FRP materials have superior corrosion resistance, they can be used as an alternative to steel in structures serving in corrosive conditions, including chemical and marine settings [7–9]. However, when subjected to severe environmental conditions, FRP composites could experience material deterioration [10–12].

In the late 1800 s, Arrhenius [13] developed a correlation model between the chemical reactions rate and the temperature. Since then, this relationship has been widely used to model the degradation of FRPs subjected to a single environmental setting [14–17]. In addition to the Arrhenius model, Park et al. (2006) [18] suggested a hyper-cuboidal volume model according to the cumulative damage approach and a stochastic process to describe material degradation because of the synergistic effect of several environmental conditions. Moisture/solution ingress is one of the most damaging factors for FRP composites [19,20]. As a result, the Fick's law [21] has received much attention and is frequently used to describe how much moisture FRP composites can absorb. Numerous models have been developed based on Fick's law to

* Corresponding author at: Faculty of Science and Technology, Charles Darwin University, Darwin, Australia.

E-mail addresses: Milad.Bazli@cdu.edu.au, M.bazli@uq.edu.au (M. Bazli).<https://doi.org/10.1016/j.conbuildmat.2023.133692>

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determine the composites' moisture content and the diffusion coefficient [22–24].

In order to examine the short-term performance of FRP composites subjected to various environments, laboratory ageing approaches are typically used to accelerate the degradation. Accelerated ageing test results have been used to extrapolate their long-term performance based on a few empirical equations. Based on this, many structural codes have proposed environmental reduction measures to manifest long-term exposure of FRPs to aggressive environments [25,26]. However, there is currently a lack of durability data for FRP materials. In addition, there are several incompatibilities between durability studies including various conditioning environments, time, temperatures, material characteristics, and testing protocols.

Three factors account for most of the difficulties in obtaining universally accurate design equations for FRP durability [27]. First, a variety of external environmental factors as well as various material compositions, such as several different fibres and resins with various fibre volume/mass ratios, affect the long-term behaviour of FRP composites [28,29]. In other words, environmental damage may progress at different rates depending on the type of fibre and matrix materials, the fibre volume fractions, and the FRP composite's geometrical configuration (e.g. plate thickness).

Second, the long-term behaviour of FRPs is affected in real life by the synergistic effect of many different environmental conditions, and this effect considerably increases the degradation process complexity. For example, alkaline water intrusion, ultraviolet (UV) radiation, wet/dry cycle, and high temperatures are all characteristics of the maritime environment, which is a typical environment for offshore structures [30,31]. Existing predictive methods cannot effectively predict the consequent synergistic effects on the mechanical properties of FRPs.

Finally, most analytical research studies on the long-term behaviour of FRPs are carried out at the microscopic molecular level, making it difficult to extrapolate the results to the macroscopic structural level [32]. The simplifications considered in the molecular model and several other assumptions used to bridge the gap between macroscopic and microscopic levels may cause the difference between molecular and structural levels. These three factors, in addition to the difficulties and shortcomings associated with conducting accelerated ageing tests, make it extremely difficult to predict the lifespan mechanical properties of FRP composites. In addition, current long-term performance predicting models proposed based on the accelerated tests in aggressive environments are scientifically overestimating the retention reductions when compared with the field data. Indeed, predicting the long-term performance (e.g. 50–100 years for a structure life span) will not be easily achieved using existing analysis and prediction methods.

It is important to note that both the Arrhenius relation and Fick's law, which are commonly used models for FRP degradation, were not originally derived specifically for this purpose. Instead, their adoption in the field is primarily due to their predictive ability based on mathematical expressions, rather than offering a complete mechanical, physical, or chemical explanation for FRP degradation.

Moreover, the accuracy of these models is heavily influenced by the calibration factors determined through experimental tests. As new test data becomes available, these models require continuous correction or modification to maintain their validity. Jiang et al. [33], in their work on predicting the stress–strain relation of FRP-confined concrete using an artificial neural network model, also highlight similar issues related to the conventional predictive models' inadequacy.

As a consequence of the limitations of predictive models, widely accepted design guides [34–39] still rely on empirically determined reduction factors. However, it is worth noting that these factors can vary between different guides and may significantly overestimate or underestimate the true durability of FRP composites. This underscores the need for further research and advancements in predictive modelling techniques to more accurately assess FRP degradation and ensure the long-term structural integrity of FRP composites [27].

The durability assessment of composite materials in different environments depends on various variables and is a highly non-linear problem [40]. However, Machine Learning (ML) techniques have shown promise in addressing this complexity issue. ML excels at handling complex non-linear structural problems, enhancing the predictability of FRP composites' behavior. By leveraging vast datasets and powerful algorithms, ML models can effectively capture the intricate relationships between influential factors and predict the long-term durability of FRP composites more accurately. Artificial intelligence (AI) has made it possible to create explicit models that are trustworthy for engineering problems [41]. Steel beams design using Artificial Neural Network (ANN) conducted by Adeli and Yeh [42] in 1989 was one of the first applications of ML in structural engineering. Since then, various AI tools have been used by researchers to address challenging engineering problems, including ANN, Adaptive Neuro-Fuzzy Inference Systems (ANFIS), Genetic Expression Programming (GEP), Genetic Algorithms (GAs), and Genetic Programming (GP) [32,43–54].

Incorporating AI models with optimising algorithms has increased the accuracy of the established models [47,48,55]. However, although progressing rapidly, the use of AI models in predicting the degradation of FRP composites under different loading and environmental conditions is in its early stages [40]. The shortage of experimental databases significantly impedes the adoption of ML in structural engineering. Such limitation becomes even more constraining when it comes to long-term performance data. Using FRP composites in construction as load-carrying structural members is relatively new, and thus structural long-term durability data relative to materials exposed to aggressive environments are limited. However, with time, more laboratory and field data are being reported thus, more reliable ML-based predictive models are being proposed by researchers.

ML methods have found valuable applications in predicting various aspects of FRP behaviour in engineering [56]. These applications include the optimisation of the manufacturing processes [57], micro-structure properties [58], accurate estimation of ultimate strength [59], durability assessment under different environmental conditions [60], prediction of fatigue behaviour [61], analysis of bond strength between FRP and concrete [62], and structural health monitoring through sensor data analysis [63]. Additionally, ML-driven optimisation aids in designing efficient FRP reinforcement configurations, while material characterisation benefits from ML techniques for predicting material properties. Despite notable progress, challenges persist, such as acquiring comprehensive and reliable data, addressing complex nonlinear behaviour, and ensuring model generalisability. Ongoing research seeks to enhance the accuracy and robustness of ML predictions for FRP applications, furthering the integration of ML into the field.

Given the abovementioned, this paper reviews the (i) current practice on durability assessment focusing on various FRP composites subjected to different loading and environmental conditions and (ii) the application of ML in enhancing the accuracy of service-life and durability models.

2. Durability and service-life of FRP composites

The most often utilised FRP composites in construction are carbon FRP (CFRP), glass FRP (GFRP), and basalt FRP (BFRP) [64]. The FRP composites have been employed in several aerospace and construction applications [65–72] because of their high strength-to-weight ratio, corrosion resistance, and suitable durability properties [23,68,73–77]. However, as noted in other studies, aggressive environments such as alkaline environments and high temperatures can degrade the mechanical properties of FRP composites [78–81]. Even though FRPs are corrosion resist, but their rate of mechanical properties deterioration increases with exposure time and conditioning temperature in aggressive environments [65,82–84].

Understanding the long-term durability of infrastructures at both the material and structural levels is crucial since civil infrastructures are

normally built for a service-life of 50–100 years [85]. So, it is vital to consider the damage mechanisms (e.g. fibre/resin interface debonding) and level of degradation (service knock down factor) of FRP composites for using these structures in critical infrastructures. Among the typical aggressive environments for FRP composites, there is moisture, alkaline environment, UV radiation, freeze/thaw cycles, and high temperature. Numerous research studies have examined the resilience of various FRP composites in such aggressive conditions [86–88].

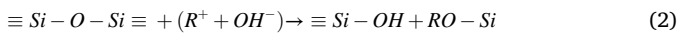
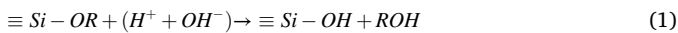
2.1. Effect of moisture

Water or excessive moisture can damage the matrix, fibres, and fibre–matrix interface [89]. Fibre rupture can occur due to the formation of microcracks on the surface of the fibres. The resin matrix is the primary component that absorbs water from the surrounding environment. The matrix commonly absorbs water via capillary and diffusion action [90]. Water ingress causes matrix swelling and microcrack progression, which influences the microstructure of the matrix [91].

Furthermore, chemical and physical degradation, such as hydrolysis and plasticisation may also occur [92]. Hydrolysis can break polymer chains and reduce the bonding force between the molecules [22]. Plasticisation softens the resin substance, lowers the glass transition temperature (T_g), and reduces the rigidity of the FRP material [93]. The resin-fibre interface plays a significant role in the water absorption effect. The fibre/resin interface is chemically heterogeneous and serves as a suitable conduit for water entry [94]. This negative effect is amplified when interfaces are of poor quality [95]. The absorbed water causes microcrack propagation and unequal interfacial swelling, leading to fibre–matrix debonding [87]. As a result, the mechanical properties deteriorate due to a reduction in interlaminar shear strength, which is induced by the degradation of the resin-fibre interface [96].

2.2. Effect of alkaline environment

It is believed that the interactions between silicate and alkali-ions in glass/basalt fibres, which cause matrix breakdown and progressive dissolution, are the primary cause of corrosion shell formation on fibres in alkaline environments [97,98]. The expansion of corrosion shells causes local stress in the network and, subsequently, debonding and exfoliating fibres at the fibre/resin interface. **Equation (1)** displays the chemical reaction that leads to the creation of the corrosion shell. The breakdown and eventual dissolution of the silicate network are shown in **Eq. (2)** [98–100].



The formed corrosion shells grow thicker with exposure time and advance into the fibre core. Due to these changes and reactions, insoluble compounds such as, iron, calcium, magnesium, titanium, and zirconium are produced [99]. As a result of the discrepancy in mechanical properties between the corrosion shell and the fibre, GFRP and BFRP composites are easier to exfoliate. Fibre pitting i.e. localized attacks is caused by the local diameter reduction of the fibres that decreases the BFRP and GFRP fibres load bearing capacity [101].

According to the literature, carbon fibres are almost resistant to most aggressive environmental conditions regardless of the environment type and exposure time. However, the resin matrix could still be damaged due to the moisture absorption, UV, etc., leading to the mechanical properties' degradation of CFRP composites.

2.3. Effect of elevated temperature

Although fibres retain their properties at moderate to high temperatures (e.g., E-glass fibres devitrify around 650 °C), the mechanical and

thermomechanical properties of polymer matrix decrease at elevated temperatures, especially near and above the glass transition temperature. When subjected to temperature fluctuations, FRP composites may experience three distinct zones over their lifecycle: (1) below T_g and higher than the highest service temperature (e.g., 60 °C); (2) near and above T_g and below the resin decomposition temperature (T_d) (around 300 °C); (3) near and above T_d , (e.g. fire condition) [78,102]. When subjected to moderate temperatures as described in (2), FRP composites soften and deform, causing a loss of strength and stiffness [103]. The glass transition temperature T_g of FRP composites typically between 65 °C and 120 °C degrees Celsius [103,104], is a temperature where the resin transforms from a glassy to a rubber-like state. Finally, exposure to high temperatures (300–500 °C) degrades the inherent structure of FRP composites, producing heat, smoke, ash, and potentially dangerous volatiles [105]. When treated to this temperature range (i.e., T_d), the resin's chemical bond, modular chains, and linkages between the fibres are all broken [106].

2.4. Effect of UV and freeze/thaw cycles

The physical and chemical reactions caused by UV radiation mainly affect only the surface of the material; hence the damage caused by UV is not as severe as that caused by other aggressive environments.

Resin stiffens at low temperatures and when exposed to freeze/thaw cycles [107]. Furthermore, in the absence of moisture, FRPs demonstrate a high resilience to freeze/thaw cycles. However, in the presence of moisture, the severity of damage to FRP composites subjected to freeze/thaw cycles increases. This is because of the freezing and consequently expansion of solution absorbed by the composite, which causes the formation of cracks and the detachment of the fibres from the matrix. However, the adverse effect of freeze/thaw cycles is reported to be less than that of water vapour and UV cycles [108].

3. Durability performance predicting models

In the past few decades, numerous studies have been focused to develop durability models for FRPs subjected to environmental conditions. For short-term durability models, different methods, such as empirical and regression-based models have been used by different researchers and, for long-term (i.e. life-span of the structure), Arrhenius-based models have been generally used [109,110]. In recent years, ML-based models were developed to better predict the short-and long-term performance of different FRP composites in different loading and environmental conditions. This section presents a short overview and a discussion about the limitations of the aforementioned methods.

3.1. Accelerated ageing test-based models

To evaluate the long-term behaviour of FRP composites using an accelerated ageing test method, Arrhenius models are commonly implemented [111]. Researchers have developed various performance predicting models based on the FRP performance in accelerated ageing tests. The following are the most typical models:

$$Y = a \log(t) + b \quad (3)$$

$$Y = 100 \exp\left(\frac{-t}{\tau}\right) \quad (4)$$

$$Y = (100 - Y_\infty) \exp\left(\frac{-t}{\tau}\right) + Y_\infty \quad (5)$$

where Y (%) is the specific mechanical property retention, Y_∞ (%) is the mechanical property retention after infinite conditioning period, t is conditioning period, a and b are regression constants, and τ is fitted parameter in Arrhenius plots. It should be noted that **Equation (3)** has some intrinsic limitations, which narrows its applicability, significantly.

Indeed, **Equation (3)** is only a phenomenological reflection of test data, with no hypothesis of the degradation process. Furthermore, it predicts that the residual strength will approach infinity at ageing time zero, which is obviously contradictory to the real test data [15]. Therefore, **Equations (4) and (5)** are more commonly used to predict the long-term mechanical behaviour of FRP composites.

Eq. (6) is used to transform the test results of a laboratory temperature to the real service (field) temperature: where T_0 is the lower temperature and T_1 is the higher temperature. In **Eq. (6)**, $\frac{E_a}{R}$ is the slope of Arrhenius plots, where E_a is the activation energy of the FRP, and R is the universal gas constant.

$$TSF = \exp\left[\frac{E_a}{R}\left(\frac{1}{T_0} - \frac{1}{T_1}\right)\right] \quad (6)$$

The selected model from **Equations (3) to (5)** is fitted to the shifted data to create master curves. Shifted data are generated simply by multiplying the exposure period of the accelerated tests results by the *TSF* of the relevant conditioning temperature.

The steps of long-term performance prediction of FRP composites exposed to solution-based conditions (e.g., saltwater and alkaline) using Arrhenius models are depicted schematically in **Fig. 1**.

Most studies in the literature determined their prediction models by generating master curves using accelerated ageing test data. However, few research studies have correlated experimental laboratory data to the field service condition data [112]. **Eq. (7)** is used to compute a field *TSF* in order to correlate the accelerated ageing test data to the actual field data: where $\frac{t_{field}}{t_{accelerated}}$ is the ratio of the time it takes the FRP composite to achieve given mechanical property retention (%) over the time it takes the composite to achieve the same retention (%) under the laboratory accelerated ageing condition.

$$TSF_{field} = \frac{t_{field}}{t_{accelerated}} \quad (7)$$

Fig. 2 schematically shows the steps to correlate experimental laboratory data to field service condition data.

3.2. Reliability-based models

The reliability method employs an acceptable failure probability, and the reliability index is utilised to assess the reliability of systems. The reliability technique may be adapted to address the time-dependent response of FRP durability in order to maintain a minimal safety level as FRP ages. In conclusion, the reliability technique aids in the refinement of FRP strength knock-down (reduction) factors, whilst the durability research (i.e. natural and accelerated ageing test data) aids in determining knock-down factors for expected lifetime and environment.

There is little literature on the reliability aspect of FRP design. It is crucial to recognise that the lack of sufficient long-term durability data for FRP structures hinders the development of a reliable framework leading to a Load and Resistance Factor Design (LRFD) approach within the context of existing deterministic FRP design codes. However, based on the limited data and rapid material and manufacturing quality improvements, critical parameters influencing FRP reliability and durability can be identified, reasonable assumptions can be made, and ultimately deterministic approaches can be integrated with a probabilistic model [113]. There is little literature on reliability approach for estimating the durability or ageing of FRP composites. Although there are several individual publications on the durability and reliability, there is a need for a systematic effort to build reliability-based durability measures for FRP composites. Karbhari et al. [114] proposed a method for allowing reliability-based qualification of novel FRP systems based on comparative criteria for the safety factor. Nowak et al. [115,116] demonstrated how reliability-based design specifications and regulations for typical building materials might be developed. Modern bridge asset management systems have also effectively employed the Markov chain and Weibull-based approaches, which use conditional probabilities between subsequent ageing stages to predict the degradation function of various structural parts (see Agrawal et al. [117]). However, the absence of a precise sample survival data is a major impediment to using the Markov chain or Weibull survival function on relatively new structures (e.g. FRP composites).

Recently, Bharil [113] has developed reliability-based models to propose a logical model framework of optimally designing FRP members for particular applications and for the desired service-life. The model

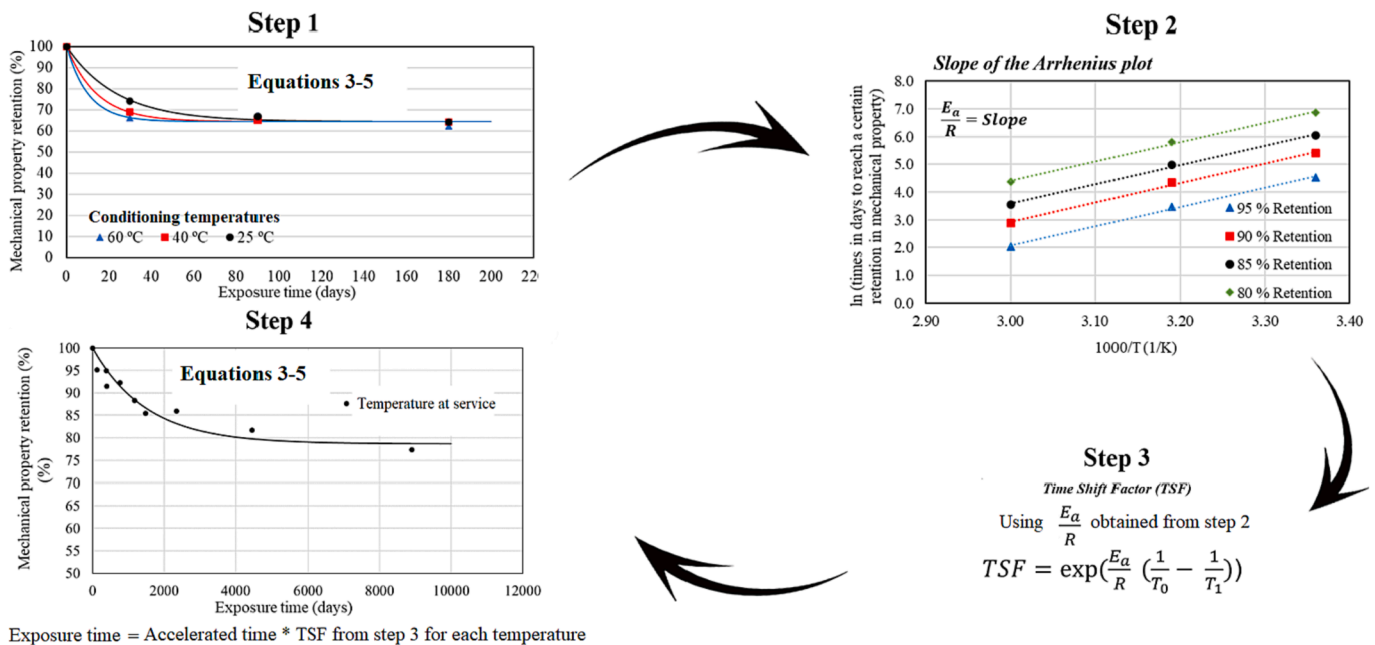


Fig. 1. Long-term performance predicting models using accelerated ageing tests: Arrhenius models.

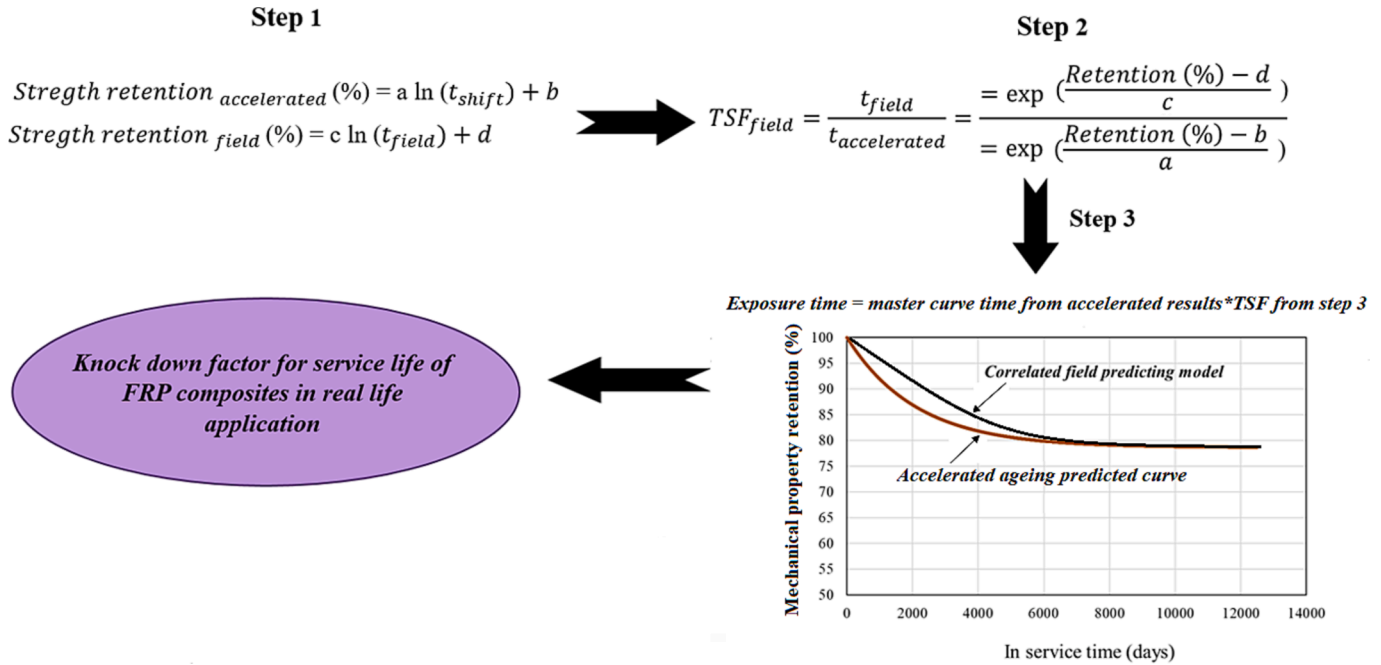
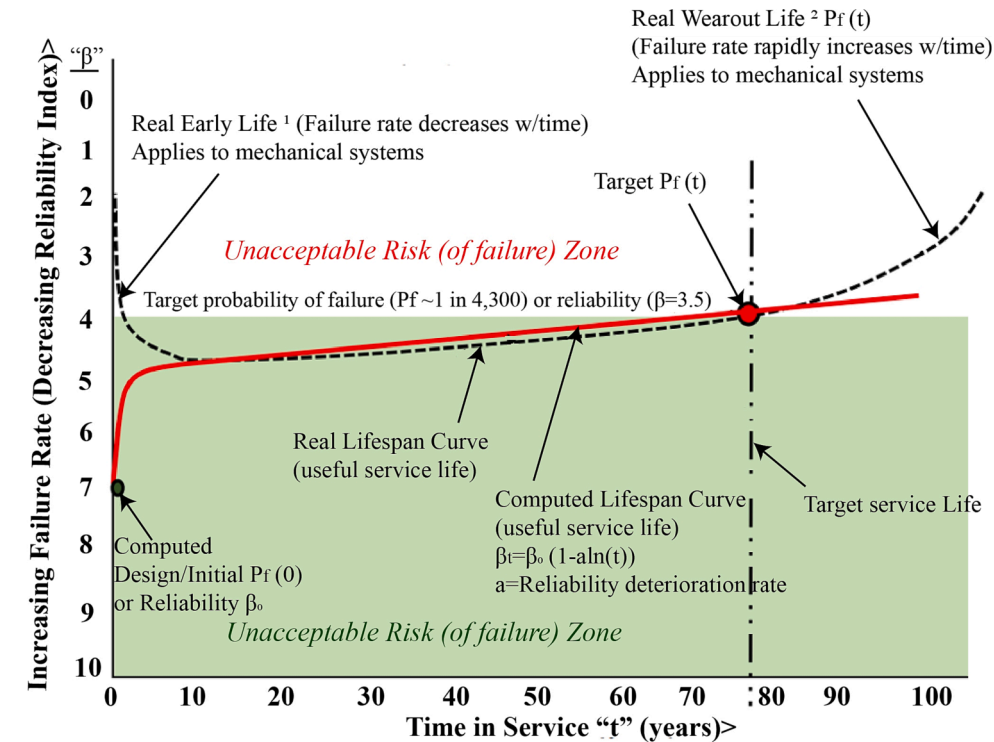


Fig. 2. Correlation between the accelerated ageing results and actual data.

considers the member's durability (i.e., time-dependent mechanical properties degradation) and criticality (e.g., importance, tolerance, ductility, and redundancy for risk), all within a range of acceptable and manageable failure risk. The method employed in Bharil's study is depicted in Fig. 3.

The dotted curve shows the behaviour seen in real-world situations, whereas the solid curve shows the reliability-age relationship as calculated. The potential of failure is initially high because of factors

including poor curing, unstable structure, poor building methods, underestimating loads, etc. The likelihood of failure, however, stabilises under-designed service circumstances once the member is put into service. After a certain amount of time, a member starts to degrade because of physical and chemical ageing; thus, during the span of its useful service-life, the member steadily loses its resistance. Using a determined reliability-durability relation, Bharil [113] closely computed/modelled this aspect of the performance. The chance of failure again rises with



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fficult to model late life

Fig. 3. Reliability-based approach used by Bharil [113] to study the durability of FRP structures.

time (often in an erratic pattern) and the level of degradation rises near the end of the member service-life. This could be due to poor maintenance, wear and tear, the accumulated combined impacts of environmental and physical causes, etc. This timeframe is depicted as a “Wear-Out” life, which is extremely challenging to model.

3.3. Machine learning based models

The invention and development of algorithms that recognise complicated patterns from experimental results, without assuming a pre-established formula as a model, and make intelligent decisions are the focus of ML, a significant branch of AI [118,119]. Models based on ML can be either predictive that predicts outcomes, descriptive that learns from the data, or both [120,121].

A limited number of design choices must be made when creating an ML model, including (i) the training type, (ii) the learning function, (iii) a representation of the function, and (iv) an algorithm to learn the function from training examples. ML is divided into supervised, semi-supervised, unsupervised, and reinforcement learning depending on the training methodology [122].

The most often ML techniques, especially in engineering, are supervised and unsupervised learning [121]. Supervised learning aims to develop a function (or model) that accurately predicts the unknown target output of upcoming instances. It starts with a training database that comprises input instances and intended outputs. The crucial component of supervised learning is that there is a “teacher” and training input–output data. Regression is the process that is used when the aim is to predict continuous target variables. However, the task is referred to as classification if the objective is to predict discrete target variables. The unsupervised learning objective is to group training examples into groups so that each group’s data exhibits a high degree of familiarity. In contrast to supervised learning, unsupervised learning lacks access to the data labels (output variable).

To solve problems using ML techniques, an algorithm must be created. Fig. 4 provides examples of some of the most important algorithms widely applied to supervised and unsupervised learning [122].

Building an ML model involves three basic steps: setting up the

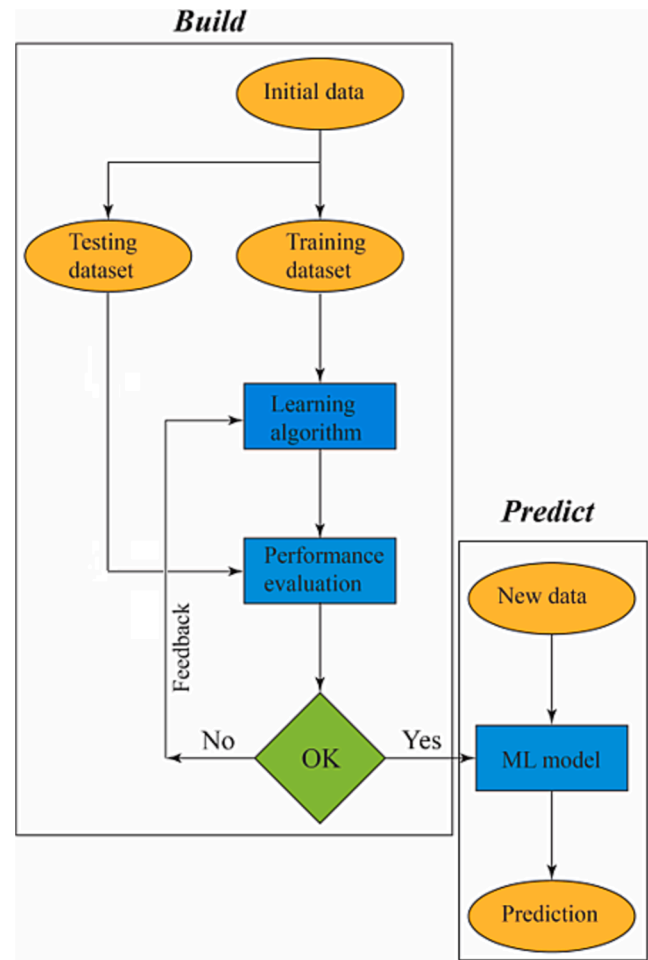


Fig. 5. Typical workflow of ML approach for prediction [123].

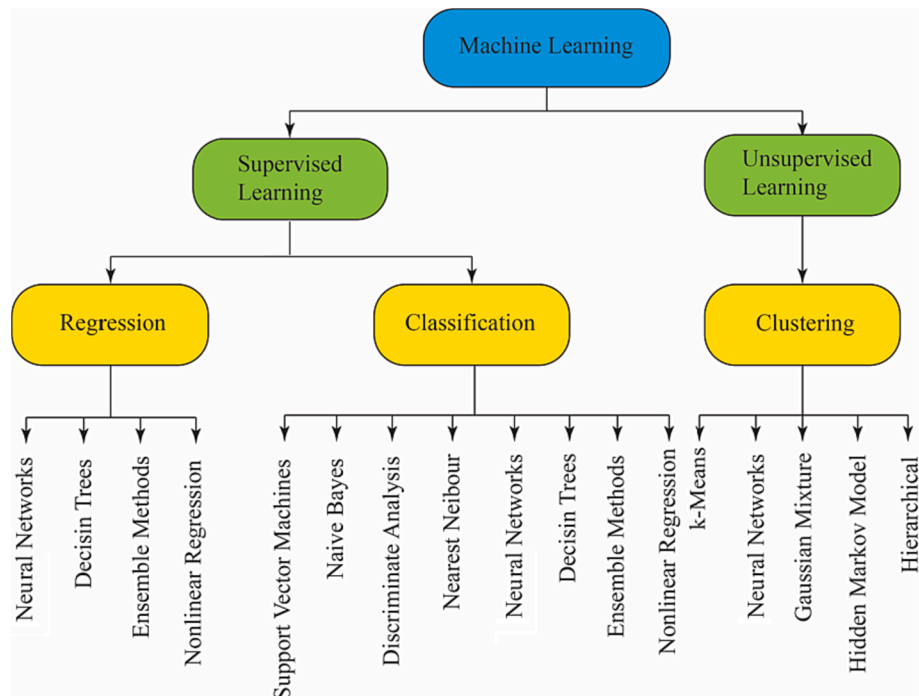


Fig. 4. Common ML algorithms used in engineering applications [122].

database, training the model, and assessing the model. Fig. 5 shows the flowchart of these steps.

With respect to the application of ML in predicting the behaviour of FRP composites, Pattnaik et al. [56] introduced a flexible procedural framework applicable from the initial selection of raw materials to the eventual performance characteristics of laboratory-scale applications. This approach, depicted in Fig. 6, empowers machine learning researchers to guide the process. The study underscores machine learning's pivotal role across various stages of polymer composite processing.

Initially, the choice of fibre, filler, and matrix aligns with end-user applications, and their compositions are cataloged as an input dataset. Machine learning algorithms optimise this composition for the final product. Subsequently, during composite fabrication, the selection of fabrication methods entails controlling parameters such as pressure, flow rate, temperature, and more. These parameters become inputs for machine learning algorithms, which identify optimal fabrication conditions, significantly reducing experimental time and costs.

Post-fabrication, machine learning remains instrumental in composite characterisation. By virtually simulating machine input parameters, this stage provides an efficient and cost-effective alternative to

physical characterisation, expanding exploration beyond the limitations of actual testing machines. Computational power aids in minimising the number of required experiments, directly impacting testing costs. Finite element modelling also establishes a mathematical framework to simulate specimen behaviour based on input parameters, with outputs compared to machine learning predictions through error analysis.

3.3.1. Machine learning applications in FRP durability prediction

The application of ML to structural engineering, specifically the application to composite materials in building, is still in its infancy. The shortage of experimental data to ensure the validation of ML models is one of the primary factors impeding the implementation of ML in predicting the performance of FRP composites at early stages [123]. Since FRP composites have been used in construction applications for a few decades, such a challenge becomes even more severe when it comes to their long-term durability.

As time passes, more experimental and numerical research data on the performance of FRP composites under different loading and environmental conditions are being reported. So, researchers can collect acceptable-sized datasets of the mechanical properties and

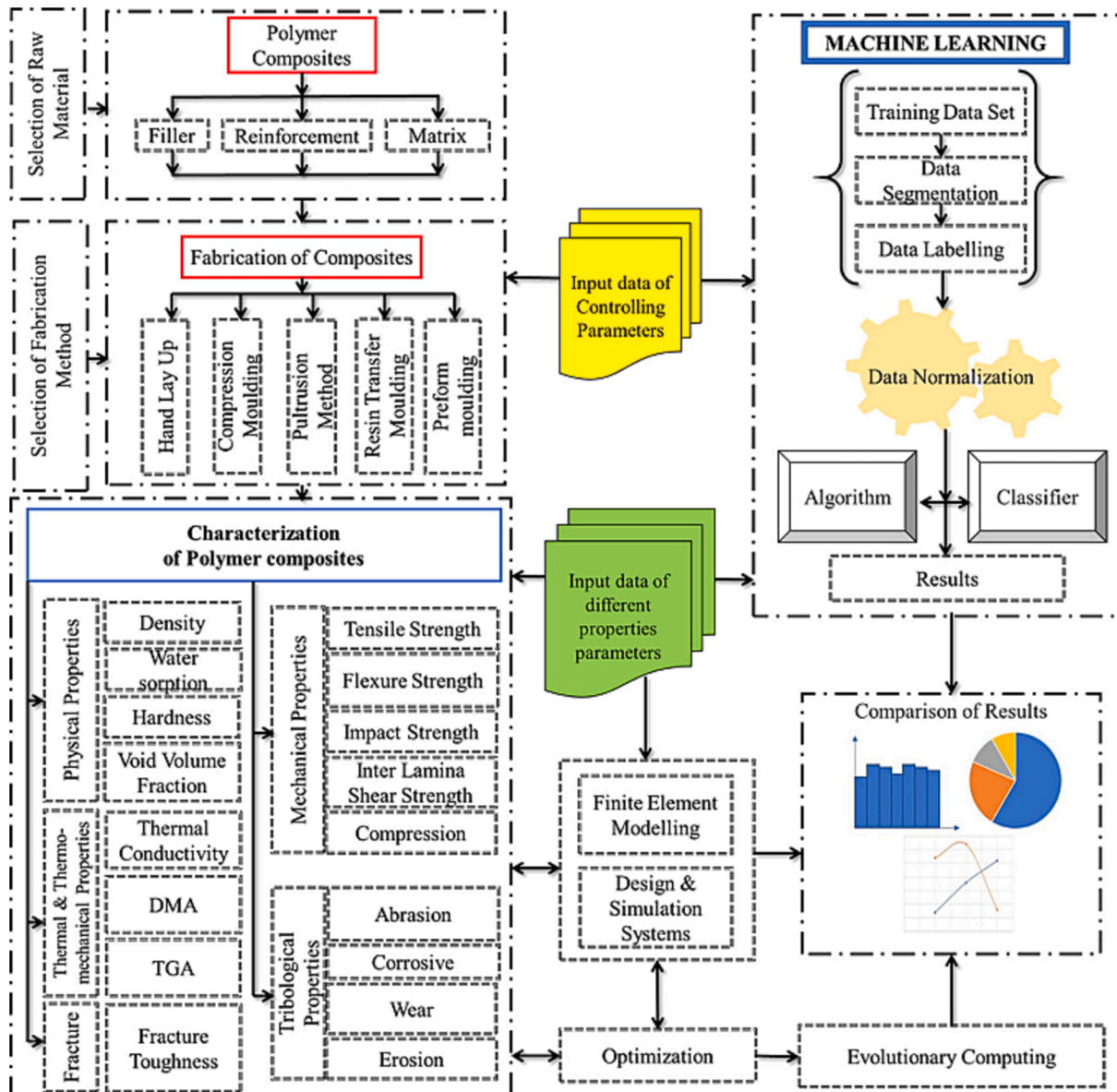


Fig. 6. Role of Machine learning in Fiber reinforced polymer composite [56]: Courtesy from Elsevier.

Table 1

Recent studies adopting different algorithms predicting the performance of FRP composites subjected to different environmental conditions and long-term loading.

Study	FRP system	Environmental condition	Property	Algorithm type	Environmental reduction factor / Predictive formula
[40]	GFRP bar	Alkaline + seawater	Tensile strength	TRF ¹	YES
[124]	GFRP bar	Alkaline + seawater	Tensile strength	GEP ² , ANN ³ , ANFIS ⁴	YES
[125]	GFRP bar	Alkaline + seawater	Tensile strength	PSO ⁵ , GA ⁶ , SVM ⁷	NO
[126]	GFRP bar	Alkaline + seawater	Tensile strength	MPMR ⁸ , DNN ⁹ , ANFIS-GA	YES
[127]	GFRP bar	Alkaline + seawater	Tensile strength	GEP	YES
[128]	GFRP Plate	Seawater	Tensile strength	ANN	NO
[27]	Various FRP composites	Water, humidity and alkaline solution	Tensile strength and modulus	XGBoost	NO
[129]	GFRP and CFRP bars and laminates	Post elevated temperature	Tensile strength and modulus	ANN, GA	YES
[130]	GFRP and CFRP bars and laminates	Elevated temperature	Tensile strength and modulus	ANN, GA	YES
[131]	GFRP and BFRP bars	Alkaline	Tensile strength	ANN	NO
[132]	GFRP plate	water	Water absorption	ANN	NO
[133]	GFRP Plate	Seawater	Acoustic emission (AE) using three-point bend	ANN	NO
[134]	BFRP Plate	Ambient	Tensile Creep	ANN	NO
[135]	NSM CFRP sheet	Elevated temperature + Sustained load	4 point bend (impedance spectra)	K-Means Clustering	NO
[136]	FRP-wrapped RC columns	Elevated temperature	Temperature variation	ANN	NO
[137]	Various FRP laminates	Moisture	FRP and concrete bond strength	LR ¹⁰ , Ridge, Lasso, DT ¹¹ , SVM, ANN	YES
[138]	GFRP Laminate	Ambient (Fatigue loading)	FRP and timber bond strength	ANN	NO
[139]	Various FRP laminates	Acidic and alkaline environment	FRP and timber bond strength	GEP	YES
[60]	GFRP bars	Alkaline environment	Tensile strength	GAM ¹² , BPNN ¹³ , ELM ¹⁴ , SVM, RF ¹⁵ , XGBoost, LSTM ¹⁶	NO
[140]	GFRP bars	Alkaline environment	Tensile strength	NR ¹⁷ , SVM, ANN, GPR ¹⁸ , CBSE ¹⁹	NO

1: Tree-based random forest; 2: Genetic expression programming; 3: Artificial neural network; 4: Adaptive neuro-fuzzy inference system; 5: Particle swarm optimization; 6: Genetic algorithms; 7: Support vector machine; 8: Probability machine regression; 9: Deep neural network; 10: Linear regression; 11: Decision Tree; 12: Generalized additive model; 13: Back propagation neural network; 14: Extreme learning machine; 15: Random forest; 16: Long short-term memory neural network; 17: Nonlinear regression analysis; 18: Gaussian process regression; 19: Combined bagging and stacking ensemble.

environmental conditions while more test data sets are needed for some specific mechanical property and environmental conditions. In this section, the studies carried out by different researchers on the use of ML to predict the mechanical properties of different FRPs under different environmental conditions are reviewed. This includes studies on bare FRP composites, bond between FRP and concrete, and bond between FRP and timber. Table 1 summarises some recent studies adopting different algorithms (mostly ML-based algorithms) to predict the performance of different FRP composites when subjected to different environmental conditions and long-term loading.

3.3.1.1. Bare FRP composites. Various studies have investigated the application of ML in modelling microstructural [58], geometrical optimisation [141], damage detection [142–144], vibration characteristics [145], water absorption [132], and mechanical performance of FRP-composites [131,146–152] under different loading and environmental conditions. However, currently, the application of these models in predicting the long-term mechanical properties under different environmental conditions is mostly limited to the tensile properties.

Zhang et al. [60] used seven machine learning algorithms, utilising input factors such as the bar's diameter, the volume fraction of E-glass fibres, the pH of the alkaline solution, the conditioning temperature, and the immersion duration, to accurately predict the tensile strength retention of GFRP bars in alkaline environment. The one-variable-at-a-time (OVAAT) technique was employed to evaluate the relative responsiveness of the five input parameters. Through this analysis, pH and temperature emerged as the two most influential factors in predicting TSR. Although robust ML models were presented, as also mentioned by Zhang et al. [60], the study had certain limitations. For instance, the analysis of the GFRP bar's TSR was based on a selection of certain influencing factors, with an emphasis on those with comprehensive data. Also, the data utilised for the study originate solely from

indoor tests, lacking representation from real-world engineering scenarios. Additionally, the study does not fully elucidate the impact of parameters on model performance. Lastly, the optimal approach for selecting model parameters remains imperfect, highlighting a potential area for further improvement.

Concerning elevated temperature, Naser [130] and Naser and Uppala [129] used ML techniques to predict the tensile properties of FRP composites (bars and laminates) under and after exposure (post-exposure) to elevated temperatures. Using an evolutionary genetic algorithm, Equations (8) and (9) were proposed by Naser [130] to predict the tensile strength and elastic modulus reduction factors (R_f and R_E) of FRP (GFRP and CFRP) composites under elevated temperatures, respectively:

$$R_f = 1.033 + 6.524e^{-15}T^2m^2 - 0.001614T - 7.735e^{-7}mT^2 - 6.077e^{-21}T^7m^3 \quad (8)$$

$$R_E = 0.9225 + \frac{1.697}{T+m} + 5.112e^{-11}(T+m)^4 + 2.88e^{-14}(T+m)^5 - 2.416e^{-8}(T+m)^3 \quad (9)$$

where T (°C) (range up to 500 °C) is the exposure temperature, and m is the material number and equals to 1 for CFRP and 2 for GFRP, respectively.

Similarly, Equations 10 and 11 were proposed by Naser and Uppala [129] to predict the tensile strength and elastic modulus reduction factors (R) of GFRP composites (bars and laminates) after exposure to elevated temperatures (post-exposure), respectively. The corresponding formulas for CFRP composites were presented in Equations (12) and (13):

$$R_f = 0.949 + 6.93e^{-14}T^5 - 4.82e^{-11}T^4 \quad (10)$$

$$R_E = 1.01 + 0.00059T - 2.47e^{-7}T^2 - 0.0208\sin(-1.03T) \quad (11)$$

$$R_{f_i} = 0.964 + 0.00208T + 7.09e^{-8}T^3 - 3.88e^{-16}T^6 - 2.8e^{-5}T^2 \quad (12)$$

$$R_E = 1.01 + 1.81e^{-16}T^6 - 80.000498T - 1.67e^{-8}T^3 \quad (13)$$

where T (°C) is the exposure temperature.

The longitudinal tensile characteristics of pultruded FRPs subjected to high humidity, water, and alkaline solutions, as the most harmful conditions for FRP composites, were studied by Liu et al. [27]. Over 700 data points related to FRP's tensile characteristics were selected from the database gathered by Liu et al. [19]. After comparing and testing several algorithms, the XGBoost algorithm was chosen in their study to construct the prediction model for the mechanical characteristics of FRP materials, taking into account the complicated degradation mechanisms of pultruded FRP composites after long-term environmental conditioning.

A variety of factors may impact the residual mechanical characteristics of FRP composites exposed to aggressive environments. The potential contributing parameters are the exposure time and temperature, fibre and matrix type, fibre volume percentage, pH value, and plate thickness.

Prediction in the field is difficult due to the synergistic effects of all these elements, which have an impact on the mechanical characteristics of FRP composites [153]. Feature impact in machine learning applications identifies the influence level of features (inputs) in a dataset on the results of a ML model. Different algorithms focus on different features in a dataset based on their properties. For example, features with strong linear trends will greatly impact linear-based methods like regression, whereas nonlinear-based methods will leverage the data's more complex relationships. For instance, an F-score has been used in a decision tree model to characterise all aspects of the prediction outcomes quantitatively. The contribution of a feature to the XGBoost decision tree can be shown by its F-score. In other words, the F-score of a specific attribute will rise as it is used more frequently in the tree decision process. Fig. 7 depicts the calculated F-scores of all features being investigated by Liu et al. [27].

Attribute importance analysis [27] has shown that the external factors of exposure time, exposure temperature, and environment pH value, as well as the internal factors of fibre volume fraction, and plate thickness, are the most significant in determining the residual tensile properties of pultruded FRP composites when they are exposed to water, high

humidity, and alkaline environments. As stated by Liu et al. [27], the developed model has the potential for ongoing updates as the dataset becomes more comprehensive. The current study specifically examines the tensile characteristics of FRP composites and considers just three environmental influences. In subsequent developments, the XGBoost model could be extended to encompass the prediction of compressive, shear, and flexural properties of FRP composites under diverse environmental conditions.

Iqbal et al., in several studies [40,124–127], predicted the tensile properties of GFRP bars under alkaline and seawater environments using various ML algorithms. It is worth mentioning that, in their studies, their dataset only covered the degradation model of E-glass GFRP bars and vinyl ester resin. To encompass the impact of different fibre and resin types, a more extensive dataset is necessary.

Using a GEP-expression trees (ETs) model, Iqbal et al. [124] proposed Equation (14) to predict the tensile strength retention $TSR(\%)$ of GFRP bars exposed to alkaline with and without seawater environment:

$$TSR(\%) = \left[\left(\frac{1}{pH - 1.89d_b} \right) \times \left(\frac{2.45t + T \times d_b}{7.68V_f} \right) \right] + \left[pH \left(\sqrt[3]{\frac{-9.36}{V_f}} + 5.64 + (7.752V_f) \right) \right] + \left[\left(\frac{3.58}{\sqrt[3]{2V_f - 2V_f}} \right) \frac{1}{pH} \right] \quad (14)$$

They considered bar diameter (d_b), fibre volume fraction (V_f), pH, conditioning temperature (T), and conditioning duration (t) as the influential factors (i.e. inputs of the model). The importance of each variable obtained in their study is presented in Table 2 for training, validation and their combination. Similar to the study of Liu et al. [27] (see Fig. 7), exposure time has the strongest influence on the tensile properties of GFRP composites. However, the effect of pH and composites thickness (i.e. diameter in Iqbal et al. and plate thickness in Liu

Table 2

Variable importance (%) in TSR (%) of GFRP bars exposed to alkaline environment [124].

Dataset	Variable importance (%)				
	d_b	V_f	pH	T	t
Training	20	26	2	25.5	26.5
Validation	18	23	3	28	28
Training and validation	26	22	3	25	25

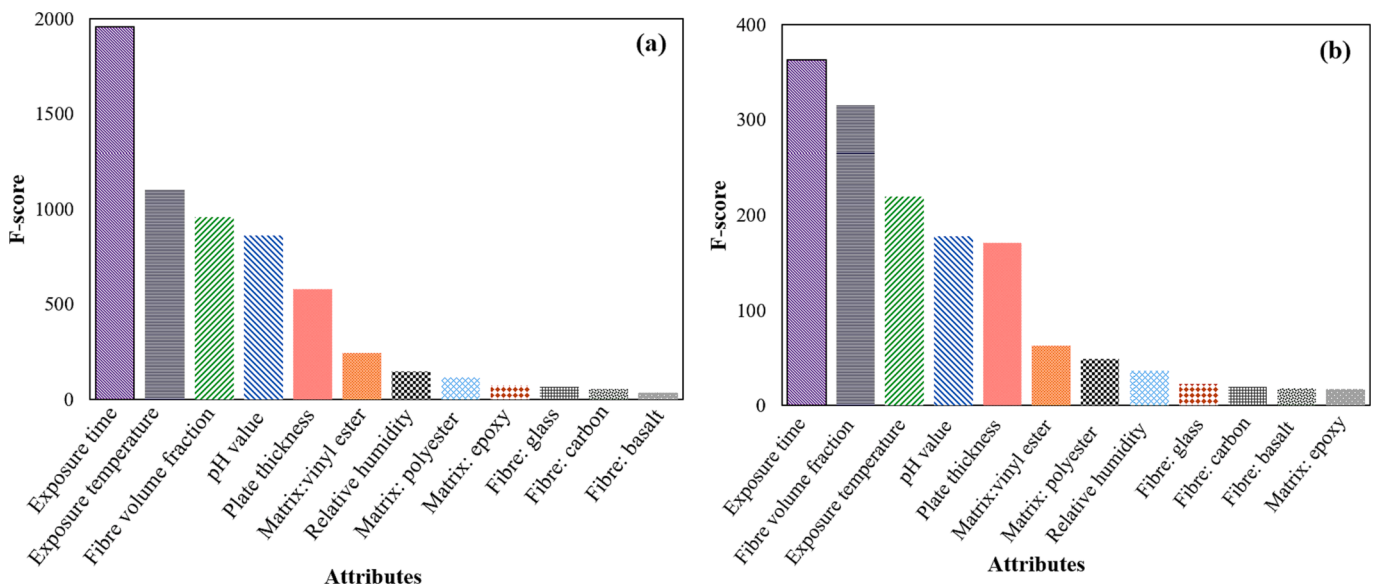


Fig. 7. Attribute importance analyses of XGBoost decision tree model: (a) For tensile strength model, (b) For tensile elastic modulus model [27].

et al. [27]) seem to have different weights on the TSR (%). Such difference could be due to the different datasets used. It is also worth mentioning that Iqbal et al. [124] only considered the alkaline and seawater environment, while Liu et al. [27] considered all water, high humidity and alkaline solution conditions in their variable importance study. It is well known that the adverse effect of alkaline environment on FRP composites mechanical properties is significantly higher than that of other environments, such as only water. This could be another reason for such a difference between the two studies.

Surprisingly, using a sensitivity analysis, in contrast to their previous study [124], Iqbal et al. [125] concluded that the influence of bar diameter, fibre volume fraction, and pH on the tensile strength retention of GFRP bars exposed to an alkaline environment are higher than the influence of exposure duration and temperature (See Table 3). In the sensitivity analysis conducted by Iqbal et al. [125] and presented in Table 3, a simulated dataset was generated, where only one of the input parameters was varied while keeping the others constant at their average values. For each variation, 10 simulated data points were recorded. The value of R_{ij} in Table 3 shows the strength of the relationship between every input variable and the output. Such contradictory conclusions could be due to various reasons, such as the quality of the data set, the accuracy of training, ML overfitting or underfitting, different ML algorithms used, etc. That is a common problem with machine learning studies in engineering; machine learning algorithms are used to investigate patterns from experimental results, but rarely relate findings back to the underlying fundamentals causing the patterns. Reviewing the literature shows that such a problem also applies when using ML in predicting the long-term durability of FRP composites when subjected to environmental conditions. Before applying machine learning algorithms in the durability performance prediction of composites, it is very important for researchers to understand the degradation mechanisms and underlying fundamental causes of damage in composites to avoid several potential challenges of using ML. For example, the nature of each variable contribution and their severity can be understood by reviewing various research studies currently available in the literature. It is known that ML output could be very promising in terms of the accuracy of the prediction, however, a poor dataset could result in the unreliable fitting. Understanding the fundamentals will help identify wrong trends and contributions in ML even if the prediction values are very close to the experimental results. A reliable database should carefully collect a wide range of information to cover many different factors, capturing the complex interaction between FRP materials and the various environmental conditions they might face. First and foremost, a strong database should have a large number of data about different kinds of FRP materials, manufacturing type, and how they will be used in structures. This diversity helps prediction models understand how these materials and structures will react when faced with different loadings in different environments.

Additionally, a reliable database should accurately track how FRPs degrade and lose mechanical performance as time goes on. Having a detailed set of data that shows how these mechanical properties reductions happen over different periods is really important for teaching ML models how FRP structures will behave in the long term. Moreover, the database should carefully record the conditions these materials are exposed to, like changes in temperature, humidity levels, chemical concentrations and reactions, and the sustained load they experience. This level of detail helps the prediction models detect the small

differences in how FRP materials degrade in different environments. Also, making sure the database has clear guidelines for how measurements were taken, how uncertainties were considered, and how quality was maintained helps the ML-based predictions be accurate and strong. Essentially, a well-explained database that predicts how FRP structures will perform over time in different environments is like the foundation of the entire process. It helps the models understand and predict the complex challenges these structures might face in real-world situations.

By using tree-based random forest model, Iqbal et al. [40] proposed knock-down environmental factors for 100-year service-life of GFRP bars exposed to concrete (alkaline) environment (Table 4).

A similar study to [124], by using a gene expression programming (GEP) tree-based model Iqbal et al. [127] proposed another model for TSR (%) of GFRP bars subjected to alkaline and seawater environment:

$$\begin{aligned} \text{TSR}(\%) = & \left[(18.28 \sqrt[3]{2V_f}) + (d_b + 62.923) \right] \\ & + \left[V_f \left(pH - (0.32d_b \sqrt[3]{d_b - pH}) \right) \right] \\ & + \left[\left(\frac{T + 4.62}{16.193 - 4.024d_b} \right) (pH - 2V_f) \right] \\ & + \left[\sqrt[3]{3V_f(T + t) - ((6.72 - t) - \left(\frac{t}{6.72} \right))} \right] \\ & + \left[\sqrt[3]{3 - \frac{2t(pH + d_b)}{34.55} + 9.48} \right] \end{aligned} \quad (15)$$

Furthermore, similar to [40], knock-down environmental factors were proposed for two GFRP types with different fibre volume fractions.

Kaloop et al. [126] compared the results obtained from an integrated adaptive neuro-fuzzy inference system with genetic algorithm (ANFIS-GA) and deep neural network (DNN) models with those obtained from the GEP model (Equation (16)) for TSR (%) of GFRP bars in alkaline environment. It was shown that based on R^2 value, the DNN and ANFIS-GA enhanced the prediction accuracy by 15.74 % and 19.10 %, respectively, compared to the GEP algorithm.

$$\begin{aligned} \text{TSR}(\%) = & t + 0.41 \left((V_f \times T - t) - \frac{V_f - T}{6.20} \right) + 0.48 \\ & + \left((1.37t - 1.56 - T) \times \left(-\frac{1.37}{2.71} \right) \right)^2 \\ & + d_b + \frac{\frac{pH}{(pH - 3.003) + (8.98V_f)}}{7.88} - d_b + T \times t \times \left(\left(-\frac{3.81}{7.37} \times t \right) - 3pH \right) \end{aligned} \quad (16)$$

Kaloop et al. [126] conducted the sensitivity analysis results of DNN and ANFIS-GA models for influential variables (Fig. 8). As is seen from Fig. 8 and comparing the results with other studies [27,124,125], it can be concluded that the contributions of different parameters may significantly vary when using different ML algorithms.

Recently, Idrisi et al. [128] examined the capability of ML in predicting the tensile strength retention of glass/epoxy composite after immersion in seawater for durations of up to 11 years at temperatures of 23 °C, 45 °C, and 65 °C. Temperature and time were considered as the only input variables in their study. Implementing Levenberg Marquardt neural network algorithm showed a high accuracy level of fitting of the experimental data to the model.

Kim and Oh [131] compared the prediction capability of multiple regression analysis (MRA), polynomial regression analysis (PRA), and an ANN for tensile strength of BFRP and GFRP bars exposed to alkaline environment. Tensile strength was considered the dependent variable, while temperature, exposure period, and bar diameter served as the independent factors. Using MRA and PRA models, it was discovered that the exposure temperature had the most significant influence on the reduction of FRPs' tensile strength, followed by exposure duration and bar diameter. Among the three algorithms, the ANN technique also had the lowest error, error rate, and the highest correlation between

Table 3

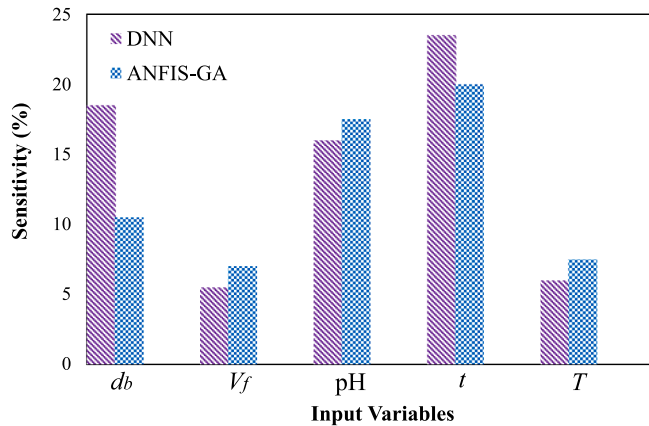
Variable importance (R_{ij}) on the TSR of GFRP bars subjected to alkaline environment [125].

Dataset	R_{ij}	V_f	pH	T	t
	d_b				
Simulated dataset	0.98	0.98	0.98	0.87	0.77

Table 4

Environmental knock-down factors for a 100-year design life of GFRP bars subjected to concrete environment.

Bar diameter (mm)	Temperature (°C)	Relative humidity (RH%)							
		100 % RH	90 % RH	80 % RH	70 % RH	60 % RH	50 % RH	40 % RH	30 % RH
9.5	20	0.72	0.84	0.89	0.92	0.95	0.96	0.98	1.00
12.7	20	0.79	0.88	0.92	0.94	0.96	0.97	0.99	1.00
15.9	20	0.77	0.87	0.91	0.94	0.96	0.97	0.99	1.00
9.5	28	0.66	0.80	0.86	0.91	0.94	0.96	0.98	1.00
12.7	28	0.75	0.85	0.90	0.93	0.95	0.97	0.98	1.00
15.9	28	0.73	0.84	0.89	0.93	0.95	0.96	0.98	1.00

**Fig. 8.** Influence of variables on the TSR of GFRP bars subjected to alkaline environment [126].

predictions and the experimental data.

Based on the time-dependent fluctuations in acoustic emission (AE) signal characteristics, Kumar et al. [133] investigated the corrosive effects of saltwater on the flexural strength of GFRP plates. Several AE factors were evaluated as input during training, including energy, absolute energy, counts, signal intensity, and impacts. The generalised regression neural network (GRNN) and the radial basis function neural network (RBFNN) models' input data were selected from 40 % to 70 % of failure loads. The RBFNN model predicted the ultimate failure strength with a percentage error of 0.5–7.2 %, confirmed by the experimental data, whereas the GRNN model predicted the ultimate failure strength with a percentage error of 0.5–4.4 %.

As is seen in Table 1, tensile properties of FRP composites, particularly GFRP composites, under different environmental condition, mostly alkaline and seawater, have been predicted using various ML-based algorithms. To the best of our knowledge the flexural strength retention was studied only in one research [133], while other mechanical properties, such as shear and compressive strengths, have not been studied. Moreover, other environmental conditions, such as UV radiation, wet/dry and freeze/thaw cycles, etc., have not been considered as well.

3.3.1.2. Bond between FRP laminate and concrete. FRP composites externally bonded to concrete components have been identified as a potentially useful method for enhancing the performance of already-built structures. The load transmission capacity of an FRP-to-concrete connection depends on the integrity of the bond between the FRP and the concrete substrate [154], which can be affected by both loading and environmental factors. Several researchers [155–164] have investigated the feasibility of using ML to simulate the bond strength of FRP-to-concrete joints. However, the developed models can only be employed using unconditioned data, therefore, they cannot be used to investigate the durability of conditioned connections [165]. Once again using ML techniques, Aghabalaie Baghaei et al. [137] assessed the durability of FRP-to-concrete connections in wet environments for the first time. A database including the results of 429 durability tests from more than 20

research was used. Predicting the FRP-to-concrete bond strength required them to investigate several ML regression approaches, including linear, decision tree, artificial neural network, ridge, lasso, and support vector machine. Failure mode classifiers were also developed using various classification techniques, including naïve Bayes, k-nearest neighbours (KNN), and ensembles. M5P, a decision tree-based model, has offered a realistic equation to predict the bond strength as a function of the mechanical, geometrical, and environmental parameters of FRP-to-concrete connections.

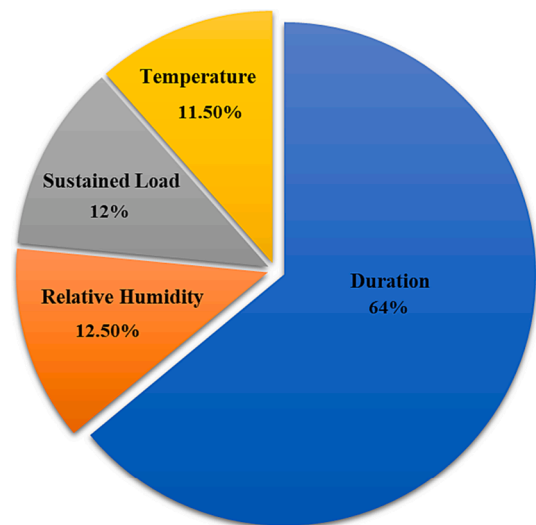
It was discovered that ANN, decision tree, and SVM give superior performance in terms of prediction compared to linear models. Moreover, the measurement values of the testing dataset suggested that, relative to other models, ANN was the most effective model in predicting unseen bond strength values. The ensemble approach has the greatest accuracy among classification algorithms with 86.2 %, followed by SVM (85.3 %), decision tree (83.7 %), KNN (83.4 %), and naïve Bayes (77.8 %).

Equation (17) was presented to estimate the bond strength between FRP composite and concrete after exposure to humidity using an M5P algorithm:

$$P_u = a + (b \times b_c) + (c \times L_f) + (d \times t_f) + (e \times b_f) + (f \times C) + (g \times T) + (h \times RH) + (i \times D) + (j \times P_r) + (k \times E_f) + (l \times f'_c) \quad (17)$$

where b_c is the concrete width, L_f is FRP bond length; b_f is FRP width, C is exposure type; T is temperature; D is the exposure duration, P_r is the sustained load; E_f is the elastic modulus of FRP, f'_c is the concrete compressive strength; a is a constant, and $b, c, d, e, f, g, h, i, j, k$, and l are the coefficients of the input variables of M5P algorithm.

The contribution of environmental factors to the identification of

**Fig. 9.** Relative contribution of environmental parameters in failure mode of FRP-concrete bond [137].

failure modes by the ensemble model is depicted in Fig. 9. It was discovered that the exposure period has the greatest impact on the mode of failure of FRP-to-concrete connections, followed by an equal contribution of relative humidity (RH%), sustained load, and temperature.

3.3.1.3. Bond between FRP laminate and timber. Although wood is widely used in construction as a sustainable material, it has several limitations, such as biological degradation over time and brittle behaviour in flexure [166]. In terms of strength and rigidity, timber-FRP composites compensate for these shortcomings [167]. Despite the well-researched bond strength predictions on the concrete-FRP interface and several practical and theoretical studies in this field, few attempts have been made to determine the bond strength between FRP and timber, especially when environmental agents are included.

Jia and Davalos [138] analysed the load ratio influences on fatigue of interfaces of phenolic FRP composites bonded to red maple wood using the ANN algorithm on a series of experimental findings. The rate of crack development as predicted by the ANN method was compared to actual results and theoretical predictions derived from a modified Paris Law equation. It was concluded that the suggested neural network model could estimate fatigue responses, such as the rate of crack formation, which would enable the establishment of design recommendations for

$$\frac{F}{E_t A_{t1}} = \left(\frac{C2 - \left(\frac{C3}{\left(\frac{S_{ad}}{S_f} - C1 \right)^2} \right)^3}{\left(\sqrt{C1 + \frac{A_f}{A_{t2}}} + \sqrt[3]{t^2} + \left(\frac{S_{ad}}{S_f} + C0 \right)^3 \right)^2} + \frac{\sqrt[3]{t \frac{E_{ad}}{E_f}} C4 \times pH \times C6^2 \times \left(\left(\frac{E_{ad}}{E_f} \right)^2 \times C5 \times t + \frac{A_f}{A_{t2}} \times C7 \right)}{(pH^2 \times C4^3)^3} \right) \quad (19)$$

$$\begin{aligned} \ln\left(\frac{F}{E_t A_{t1}}\right) = & \frac{\ln\left(\frac{A_f}{A_{t2}}\right) \ln\left(\frac{S_{ad}}{S_f}\right)}{(\ln(pH))^3 \left(\ln\left(\frac{E_{ad}}{E_f}\right) \ln(pH) - \ln(pH) \ln\left(\frac{S_{ad}}{S_f}\right) \right) - (C3 - C1) + (C0 + \ln(pH))} + \ln(t) \left(\ln\left(\frac{A_f}{A_{t2}}\right) \right)^2 \left(\ln\left(\frac{S_{ad}}{S_f}\right) \right. \\ & \left. - C4 \right) + \frac{\ln(t)}{\sqrt[3]{\left(\ln\left(\frac{E_{ad}}{E_f}\right) - \ln(pH) \right) \left(\ln\left(\frac{E_{ad}}{E_f}\right) C6 + \ln(pH) C5 \right)}} \quad (20) \end{aligned}$$

hybrid material connected interfaces.

Recently, Palizi and Toufgh [139] utilised a set of test data of the pull-out experiments on the FRP-timber interface obtained for two cases: ambient environment (no environmental impacts) and alkaline and acidic solutions. A supervised GEP was used to estimate the bond strength of the timber-FRP interface in ambient conditions and the strength loss in acidic and alkaline environments. Statistical evaluations were used to suggest and validate the formulations.

Equations (18)–(20) were proposed, respectively for the FRP-to-timber bond strength prediction in cases of (i) no environmental condition; (ii) short-term exposure to alkaline and acidic environment; and (iii) long-term (up to 12 months) exposure to alkaline and acidic environment.

$$\begin{aligned} \frac{F}{E_t A_t} = & \left[\left(\left(\frac{E_{ad}}{E_f} \right)^6 (C4 \times C0 \times C1) \frac{S_{ad}}{S_f} \left(\left(\frac{S_{ad}}{S_f} - C2 \right)^3 \right. \right. \right. \\ & \left. \left. - \sqrt{C3 \frac{E_{ad}}{E_f}} \right) \right) + \left(\left(\left(C7 + \frac{A_{ad}}{A_f} + \frac{E_{ad}}{E_f} \frac{A_{ad}}{A_f} \right) \sqrt{\frac{E_{ad}}{E_f}} \right) + \left(\left(\frac{S_{ad}}{S_f} \right. \right. \right. \\ & \left. \left. - C8 \right) \times C5 \times C8 \right) \left(C6^3 + \left(\frac{S_{ad}}{S_f} \right)^2 \right)^3 \left. \right] \quad (18) \end{aligned}$$

Where E_{ad} , E_f , and E_t are the adhesive, FRP, and timber elasticity modulus, respectively. A_{ad} , A_f , and A_t are the adhesive, FRP, and timber areas, respectively. A_{t1} , and A_{t2} are the timber transverse cross-section area, and timber cross-section parallel to the FRP, respectively. S_{ad} , and S_f represent the adhesive and FRP tensile strength respectively. t is the elapsed time in days and C0–C8 are constants of the predicted equation as per [139].

4. Recommendations for future studies

ML applications have recently had great success across a variety of composite subjects. However, further efforts are required to promote wider applications of ML for long-term performance prediction of FRP composites performance when exposed to various loading and environmental conditions. Some of the major future studies recommendations could be summarised as follows:

1. Creating a dataset platform for the durability of FRP composites in order to properly train ML models: Although efforts to create database platforms have begun, a specialised platform for the durability of FRP composites is required. The database could be classified based on the composite mechanical properties, e.g. tensile, compressive, flexural, shear, etc. and the environmental conditions, e.g. alkaline, acidic, seawater, moisture, UV, wet/dry and freeze/thaw cycles, etc.

2. Using ML for long-term mechanical properties of FRP composites other than tensile strength: it is well known that the most strength degradation of FRP composites, when subjected to aggressive environmental conditions, are associated with interlaminar shear strength degradation [168]. Compressive strength was also found to be more

vulnerable than tensile strength. However, most ML-based durability prediction of FRP composites is limited to tensile strength. Future studies could be focused on other mechanical properties considering a fair number of experimental data available in the literature, particularly on the interlaminar shear strength.

3. Using ML for long-term mechanical properties of FRP composites for different environmental conditions: currently, the main focus of the literature on the durability of FRP composites is related to alkaline, seawater and moisture (water). However, ML algorithms' efficiency in predicting the mechanical properties retention of FRP composites under harsh daily environmental conditions, including UV radiation, fire, wet/dry and freeze/thaw cycles, has not been investigated. This could be a challenge considering the scattered nature of the available data reported by different researchers.

4. Detailed comparison between the knock-down environmental factors derived from conventional time-temperature based predictive models, e.g. Fick's law and Arrhenius theory, with those derived from ML-based models: several existing models in the literature derived from accelerated ageing tests show degradation to converge to a particular value after a specific exposure time. However, if short-term accelerated testing data that do not show signs of convergence at early stages is used as the datasets for ML-based models, the extrapolated output for a long-term service period, e.g. 100 years, may not be realistic. Therefore, comparing the two methods for the extrapolation range is necessary.

5. Comparative studies between different ML algorithms: Due to the wide range of methods available, each with its pros and cons, selecting the optimum ML algorithm is challenging. This is particularly true for researchers in the structural engineering community because of their likely relatively less knowledge of ML. Various ML algorithms have been adopted in the performance prediction of FRP composites under different loading and environments. However, sensitivity analysis showed that different variables (e.g. exposure time and temperature, pH, fibre volume fraction, etc.) influence weights when using different algorithms. Therefore, comparative studies are needed to investigate the efficiency of different algorithms. The analysis should be conducted not only based on the statistical performance but also the background knowledge of FRP degradation mechanisms.

6. Enhancing the durability dataset: it is essential to emphasise the need for conducting additional studies focusing on the long-term performance of FRP composites at the structural level. These studies would pave the way for developing reliability-based models and calibrations of reduction factors, taking into account both material and structural perspectives under various environmental and load conditions. Furthermore, the availability and accessibility of databases play a critical role in influencing both reliability-based models and machine learning techniques. To optimise research efforts, it is advisable to leverage existing databases for the intended purposes. Additionally, ML techniques can be employed effectively to conduct reliability analysis, presenting a promising avenue for future investigations in this field.

5. Summary and conclusion

Researchers have proposed several different ML techniques for creating quantitative predictions about the durability of composites using test and simulation data. The ML approach has proven to be effective in predicting the residual properties, especially tensile properties of FRP composites exposed to different environmental conditions, such as elevated temperature, water, high humidity, seawater and alkaline solution. However, the quality and comprehensiveness of the database, as well as the choice of the algorithm, play an essential role in prediction accuracy. Although all algorithms used by different researchers seem to predict the mechanical properties retention of FRP composites with a relatively small error margin, sensitivity analysis shows different influence contribution of variables if different dataset or ML algorithms are used. This could be due to various factors, such as poor-quality/insufficient dataset, poor-quality/insufficient training,

overfitting and other factors. Therefore, this suggests the need for a more careful use of ML in long-term durability studies of FRP composites. It is strongly recommended to pay a great attention to the knowledge background behind the degradation mechanisms of FRPs in each environmental condition and the reason why and how each variable contributes to the degradation of such material. Considering such background knowledge will help to understand the efficiency of the algorithm used, avoid overfitting, and help in the data selection process, avoiding problems due to shortage and scattering of the data.

The accuracy, efficiency, and effectiveness of ML approach change with the quality of the database. More experimental data, especially for flexural, compression, and shear (interlaminar shear), are needed to improve the current databases available in the literature so researchers can effectively apply ML in more applications for FRP composites under different loading and environmental conditions.

Finally, the efficiency of ML algorithms and the quality of training datasets greatly influence the accuracy and reliability of ML based prediction models. That is why many structural engineers and designers are still hesitant about using ML techniques for analysis and design purposes. Given that structural safety is a top priority in structural analysis and design projects, ML-based prediction models should be considered with caution, even by those who are well familiar with the subject.

CRediT authorship contribution statement

Chiara Machello: Methodology, Resources, Writing – original draft. **Milad Bazli:** Conceptualization, Methodology, Project administration, Resources, Supervision, Writing – review & editing. **Ali Rajabipour:** Methodology, Resources, Supervision, Writing – review & editing. **Hooman Mahdizadeh Rad:** Methodology, Resources, Supervision, Writing – Review & editing. **Mehrdad Arashpour:** Methodology, Resources, Writing – review & editing. **Ali Hadigheh:** Methodology, Resources, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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